DORAHACKS · PORTALDOT HACKER HOUSE 2026 · FINALIST

AgentTab on Portaldot V3

Pay-per-call escrow tabs for AI agents — open once, voucher every call, settle in batch. The same AgentTab is reached from the EVM RPC and from Substrate Revive.

Track: Modular Cloud & Middleware | Chain 420420777 | BUIDL #50150 | Gateway 402 live

Repo: github.com/zwang1111/agenttab-portaldot · BUIDL: dorahacks.io/buidl/50150

THE PROBLEM

AI agents can pay. They can’t manage API keys.

Human-shaped billing breaks agents

- Subscriptions and dashboards assume a human operator
- API keys are secrets agents shouldn’t hold long-term
- Per-call on-chain gas kills micropayments

What machine-to-machine needs

- Escrow once, pay incrementally
- Cryptographic vouchers, not passwords
- Provider batch settlement without trusting the agent after deposit

WHY PORTALDOT V3

EVM apps on a Substrate-native stack

Execution

Portaldot V3 exposes an EVM-compatible layer (pallet-revive). Solidity contracts ship with familiar tooling — Foundry, cast, viem — while staying on Portaldot's testnet.

Native tPOT tabs

AgentTab escrows native **tPOTv3** (14 decimals on Substrate) or any ERC-20. Middleware for agent payments fits the Modular Cloud track.

Substrate path, same contract

`ReviveApi_call` from the deployer fallback account matches `eth_call` for `remaining(1)`. Unsigned `Revive.call` bytes are recorded. No signed extrinsic yet.

Network facts

Chain ID `420420777`
RPC `test-evm-rpc-v3.portaldot.dev`
Console node-console.portaldot.dev

SOLUTION

AgentTab — escrow + EIP-712 vouchers

Agents escrow tPOT once, sign a voucher per API call; providers batch-settle on Portaldot V3 — micropayments without API keys or per-call gas.

open

Payer deposits native tPOT (or ERC-20) for a provider, with expiry. One tab, one escrow.

voucher

Each call: agent signs EIP-712 `(tabId, cumulative)`. The live gateway returns HTTP 402 when the voucher is missing. Tab #2 was opened and settled on testnet.

settle

Provider submits cumulative + sig once for N calls. Contract pays `cumulative - claimed`.

refund

After expiry, payer reclaims unspent balance. Reentrancy guard + existence checks on every path.

ARCHITECTURE

Agent → Provider → AgentTab on Portaldot V3

01
AI agent
open(provider, NATIVE, deposit, expiry) with tPOT value

→

02
API provider
request + voucher sig; respond or HTTP 402

→

03
AgentTab.sol
settle batches N calls; refund after expiry

Status (2026-10-29, Asia/Shanghai)

Gateway + SDK live (tab #2 settle). Dashboard reads the public RPC. Substrate `ReviveApi` matches `eth_call`. Reuse: author's MIT AgentTab (Celo, 2026-10-15) — this repo does not ship that x402 facilitator. `forge test` 7/7. Native tabs, `call{value:}` payouts, reentrancy guard.

LIVE TESTNET PROOF

Deployed on Portaldot V3 · chain 420420777

Contract	`0x9a096C216f79e08C2DB95d92F412697D8F1f9329`
Deploy tx	`0xe8a4376602a5467574dd7f2e966006da4fe41275d28a989e33f116fa57baab72`
Demo open	`0xc27ab271b49103a257046c7458805ed1c73988d464f61aa98bfe6338a13e534e`
Demo settle	`0xd58678d04863861cda814284eb2879d0abcaf85ca64779cc5e69f2a670b8ac5b`
Deployer	`0x9e2e2024c7dcA86e57608e2ceAD160081c381fDd`
Explorer	node-console.portaldot.dev · see also `DEPLOYMENT.md`

Tab #1: deposit 1e16 wei, settle 5e15. Gateway tab #2 open 0x9922d9a7… / settle 0x2031c40d…. Full hashes in DEPLOYMENT.md.

DEMO FLOW

open → voucher → settle (on-chain today)

OPEN
Tab #1
payer = deployer
provider = 0x7F22...3369
deposit = 1e16 wei native

→

VOUCHER
EIP-712
agent signs (tabId=1, cumulative=5e15)
off-chain, zero gas

→

SETTLE
Provider tx
pays 5e15 to provider
remaining(1) == 5e15
hash 0xd58678d0...ac5b

Reproduce locally

`npm run dry-run` · `npm run dashboard`

402 needs no key. Dashboard reads the RPC. Hashes: `DEPLOYMENT.md`. Runbook: `docs/FINAL-DEMO.md`. `demo-checkin.mp4` is static cards, not this demo.

THROUGH THE 2026-11-06 FREEZE

Done, and what is still open

Gateway + SDK — live

Unpaid `GET /quote` is HTTP 402. Tab #2 open and settle are on testnet. SDK speaks chain 420420777 and the 14/18 decimal split.

Substrate — probed

`ReviveApi_call` matches `eth_call`. A signed `Revive.call` extrinsic is still optional, not submitted.

Dashboard — on main

Static inspector reads the public RPC. github.io was HTTP 404 on 2026-10-29; Pages is not enabled yet. Local: `npm run dashboard`.

Still open

A real product demo video (check-in mp4 is static cards). DoraHacks final form is user-login. Freeze 2026-11-06 12:00 Asia/Shanghai.

ASK

What finalists still need

Before 2026-11-06 12:00 Asia/Shanghai:

1) A hosted dashboard URL — Pages is configured in-repo, github.io was still 404
2) Optional review of a signed `Revive.call` (dry-run already matches)
3) The DoraHacks final form is user-login — the entrant submits it, not CI

Gateway settle and tab #1 are already on-chain. Reuse: MIT AgentTab, Celo base, no x402 rail here.
github.com/zwang1111/agenttab-portaldot · BUIDL 50150

Thank you — AgentTab × Portaldot